

Research papers

Ecological drought is globally driven by soil characteristics rather than climate aridity

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ABSTRACT

Although the distribution of vegetation growth is largely determined by soil moisture availability, what determines the use of soil water at different depths by vegetation still requires investigation. Here, we used remote sensing proxy of global vegetation growth and soil moisture data to link gridded layered soil water use patterns with climate aridity, soil texture, root depth and soil water distribution characteristics. We found no significant differences in the soil water use patterns between the climate subtypes. Soil texture can explain the divergence in soil water dependence between the surface and subsurface soil layers by regulating both the layered soil water distribution and root water uptake depth. However, the controlling effect of a coarser soil texture to enhance deepwater uptake is limited in water-limited regions. A structural equation model reveals the mechanisms of vegetation growth dependence on surface and subsurface soil moisture globally. Layered soil water distribution regulated by soil texture can determine whether vegetation water stress is mostly dominated by the variation in surface or subsurface soil moisture supply across climates, and this effect peaks in global semiarid regions. Our results suggested that soil characteristics, rather than climate aridity, dominantly regulate global soil water use for vegetation growth, which is crucial for understanding the occurrence of ecological drought as a new circumstance in simulations of global change.

1. Introduction

As water availability is fundamental for plant growth, terrestrial ecosystem at the global scale is expected to become more water constrained with ongoing and warmer climate change (Jiao et al., 2021; Li et al., 2022a). Previous researchers who explored dominant hydrometeorological controls of vegetation growth have recognized that vegetation growth is significantly impacted or even dominated by soil water supply capacity across different ecosystems at local, regional and global scales compared with climatic aridity (Li et al., 2022b; Liu et al., 2023; Zhang et al., 2023; Kannenberg et al., 2024). Similar to climatic aridity, soil moisture has horizontal heterogeneity and temporal variability, but the greatest difference from climatic aridity is that soil moisture is also characterized by interlayer variability (Li et al., 2017; Jia et al., 2020; Datta and Faroughi, 2023), as unpredictable rainfall pulses need to be translated into usable soil moisture pulses in soil layers to support

vegetation growth (Crausbay et al., 2017; Slette et al., 2019; Crausbay et al., 2020). The question of particular interest is whether separation of soil water into different layers with depth occurs to support vegetation growth and its determinants. Therefore, defining and quantifying the layered soil water use pattern is essential for advancing our understanding and prediction of ecological drought.

From the perspective of plant soil water use, a more accurate representation of the vegetation response to soil water stress requires the differentiation between soil layers. Plant water uptake often occurs in two soil layers, one in shallow soil, where water is available in pulses, and the other in deeper soil, where water availability is assumed to be more constant (Schwinning and Ehleringer, 2001; Zhu et al., 2016). Currently, soil moisture datasets at the regional and global scales are available both spatially and temporally, as global observations of surface soil moisture are now available from Earth-orbiting satellites (McColl et al., 2017; Feldman et al., 2023; Zhang et al., 2024). Comprehensive

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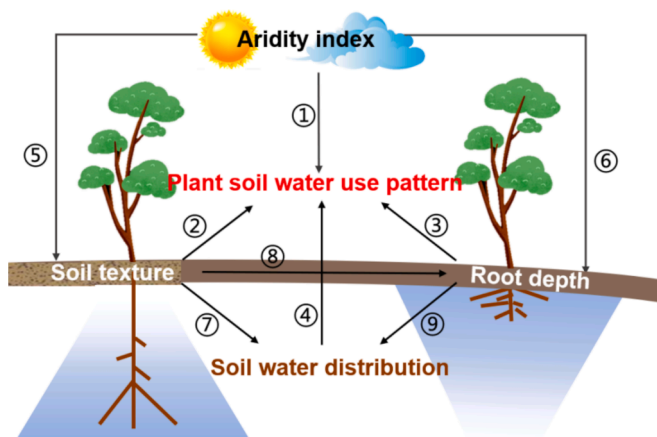


Fig. 1. Illustration of complex plausible interaction pathways derived directly from previous studies in which the major factors were hypothesized to regulate global soil moisture accessibility and availability for vegetation growth in different soil layers, including the direct effects of climate aridity (Path 1), soil texture (Path 2), root depth (Path 3) and soil water distribution (Path 4) characteristics on plant soil water use patterns and plausible indirect effects, such as climate effects on soil texture (Path 5) and root depth (Path 6), soil texture effects on soil water distribution (Path 7) and root depth (Path 8) and the effects of root-water uptake depth on soil water distribution (Path 9).

analyses of vegetation–soil water relations can benefit from a state-of-the-art soil moisture reanalysis dataset covering layered soil moisture (Martens et al., 2017). To our knowledge, a comprehensive assessment of the soil layer effect using full satellite records of long-term vegetation and soil moisture data to separate surface and subsurface soil moisture dependence is lacking, and whether the surface soil layer or subsurface soil layer determines ecosystem sensitivity to soil moisture variability is not yet understood globally. Quantifying and understanding the determinants resulting in diverse sensitivity patterns across soil layers are fundamental for inferring ecosystem vulnerability.

Observational evidence based on in situ multidepth soil moisture measurements demonstrates that soil characteristics control surface and subsurface soil moisture availability for vegetation growth, possibly by regulating layered soil water distribution or root depth distribution (Ding et al., 2020; Li et al., 2022c); for example, coarse-textured soils not only enable deep infiltration but also encourage enhanced deepwater uptake by roots (Schenk and Jackson, 2005; Sala et al., 2015; Fan et al., 2017; Jiang et al., 2022). Moreover, a key finding from previous works also indicates that the same species/age also displays contrasting root

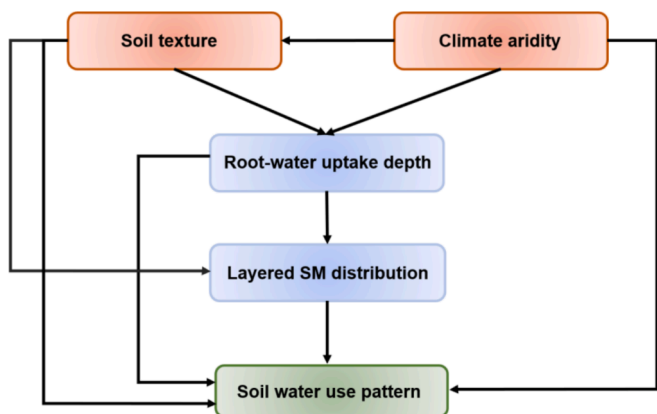


Fig. 2. A structural equation model (SEM) starts with all plausible interaction pathways derived directly from our theory that climate aridity, soil texture, root-water uptake depth and soil water distribution are all hypothesized to regulate soil water use patterns.

depths in monoculture crops and plantations, as roots are highly adaptable to local soil characteristics (Follett et al., 1974; Hodge, 2004). Therefore, it is reasonable to hypothesize that soil characteristics are the dominant factors affecting vegetation soil water use patterns at local and site scales. At regional and global scales, however, climate aridity, soil texture, root depth and soil water distribution interact with each other, so how these characteristics jointly determine layered soil water use patterns for global vegetation remains to be tested (Fig. 1).

In this study, we depicted the relationships between layered soil moisture and vegetation growth using a remote sensing proxy of global vegetation growth (i.e., NDVI) and soil moisture data to advance the understanding of the layered soil water use pattern globally. We also investigated how this soil water use pattern is modulated by soil layers with depth-varying hydrological properties by establishing a structural equation model to link climate aridity, soil texture, root depth and soil water distribution, which can reveal the critical paths regulating ecological drought sensitivity across soil layers. We aimed to obtain a more comprehensive assessment of soil moisture effects across layers on vegetation growth, which contributes to understanding the determinants and mechanisms of layered soil moisture availability for vegetation growth as a new focus of ecological drought research.

2. Materials and methods

2.1. Climate, soil and vegetation dataset

The data used in this study were openly available in the following databases. The normalized difference vegetation index (NDVI) dataset was obtained from third-generation biweekly Advanced Very High Resolution Radiometer (AVHRR) data from the Global Inventory Modeling and Mapping Studies (GIMMS) project as a proxy for vegetation growth between 1982 and 2015 (Tian et al., 2015). The global soil texture data were extracted from version 1.2 of the Harmonized World Soil Database (HWSD) (available at <http://www.fao.org/soils-portal/soil-survey/soil-maps-and-databases/harmonized-world-soil-database-v12/en/>), which represents the most comprehensive soil database to date. The dataset combined existing regional and national updates of soil information worldwide and provided global soil texture at a resolution of 30 arc-seconds based on the U.S. classification standard for sand, silt, and clay fractions (Kochy et al., 2015). The global dataset of necessary root-water uptake depths provided an estimated maximum depth of root uptake averaged over ten years (2004–2013) with global grids of 1 km (available at <https://wci.earth2observe.eu/thredds/catalog/usc/root-depth/catalog.html>) (Fan et al., 2017). The land cover data come from the EOS/MODIS data of NASA, available at <https://lpdaac.usgs.gov/products/mcd12q1v061/>. In this study, the land cover maps were reclassified, and three dominant categories (water bodies, cropland, urban and built-up land) were eliminated based on the original 17 land use types of the International Geosphere-Biosphere Program classification scheme. Climate aridity was quantified by the most widely used aridity index (AI), which represents the balance between precipitation supply and atmospheric water demand (potential evapotranspiration, PET). The generalized climate classification schemes for the aridity index values are 0.2, 0.5 and 0.65, corresponding to the thresholds for global arid, semiarid and subhumid regions, respectively (Zomer et al., 2022).

2.2. Selection of the soil moisture dataset

In this study, we intended to distinguish the contribution of two soil layers to vegetation growth: a pulse-dominated surface soil layer and a deeper subsurface soil layer that is continuous, stored and generally has less saturated soil moisture. The two soil layers set in Global Land Evaporation Amsterdam Model (GLEAM) and the corresponding algorithms are most in line with the two-layer pattern setting of vegetation soil water use proposed in this study within the range of available soil

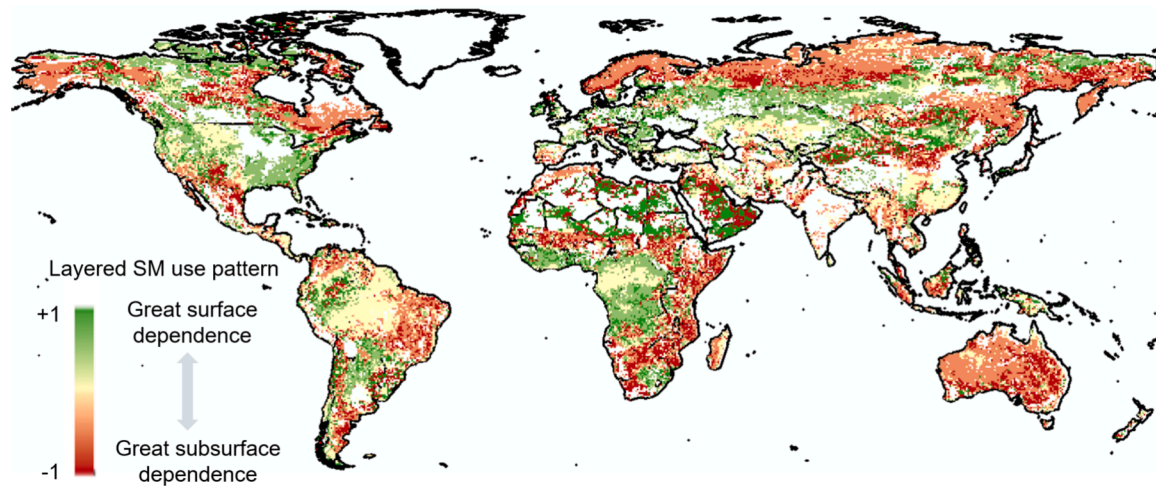


Fig. 3. The global layered soil water use pattern for vegetation growth. The spatial distribution of the quantified soil water use pattern is represented by the dependence on surface and subsurface soil moisture for vegetation growth; the closer the value is to -1 , the greater the dependence of vegetation growth on subsurface soil water supply, and the closer the value is to 1 , the greater the dependence of vegetation growth on surface soil water supply.

moisture data. There are two main reasons for this.

Firstly, the surface and subsurface soil moisture are expected to be independent output, rather than unified output by the same multi-layer water balance model. Many reanalysis datasets that cover layered soil moisture, such as the Global Land Data Assimilation System (GLDAS) and Center for Medium-Range Weather Forecasts (ERA5), their layered soil moisture data are all derived from soil module outputs. While the surface soil moisture from GLEAM is measured via satellite-driven observations, and subsurface soil moisture is estimated by precipitation minus evapotranspiration, which is closer to the two different layers of surface pulse water supply and deeper storage water supply that we proposed, and offers a greater level of independence between surface and subsurface soil moisture than other reanalysis data products, such as

GLDAS and ERA5 (Yang et al., 2022).

Next, the selected dataset should not consider root dynamics and layered soil water use strategies for vegetation in soil moisture estimation. Different from the GLDAS and other reanalysis data products driven by ecological process models, GLEAM is a semiempirical model that uses root zone soil moisture to directly constrain potential evaporation rather than setting root dynamics and soil water use at different depths, as in many ecological process models.

Based on the above two points, the terrestrial soil moisture from the GLEAM is likely a good choice.

Therefore, the GLEAM dataset was selected for this study, and the terrestrial surface and subsurface soil moisture data were downloaded from GLEAM v3.7a (available at <https://www.gleam.eu/>), which

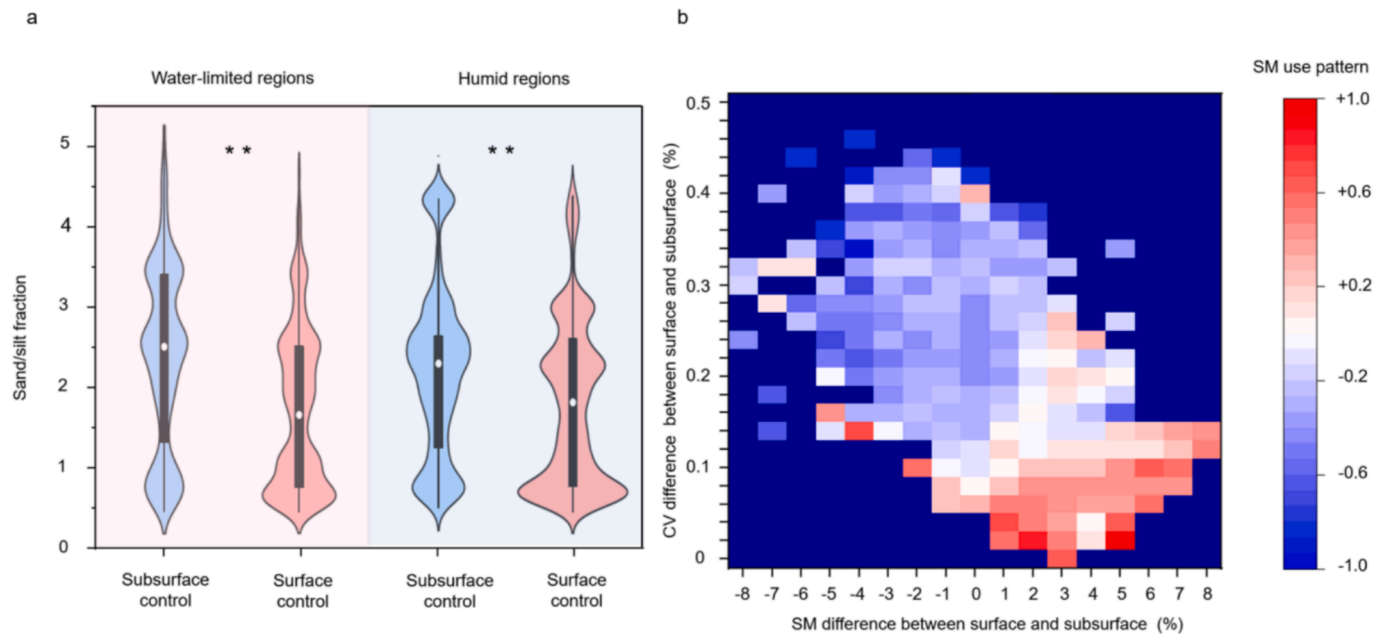


Fig. 4. Comparison of soil texture characteristics represented by the ratio of the sand/silt fraction for different soil water use patterns (a). Boxplots show the median (the white plot in the box) and the 25th and 75th percentiles (top and bottom of the box, respectively), and ** represents P values less than 0.01. The relationships between the layered soil water distribution characteristics and the quantified soil water use pattern (b). The relationships between the soil water use pattern (the closer the value is to -1 , the greater the dependence of vegetation growth on subsurface soil water supply, and the closer the value is to 1 , the greater the dependence of vegetation growth on surface soil water supply) and soil water distribution characteristics represented by the soil moisture difference between surface and subsurface; the difference in soil moisture stability between surface and subsurface are quantified by the coefficient of variation (CV).

Table 1

Set of hypotheses and related predictor variables used for analyzing the soil water use pattern.

Variable	Quantitative method	Description
Soil texture	Soil sand/silt ratio	Coarser soil texture was a higher soil sand fraction and a lower silt fraction
Climate aridity	Aridity index (Precipitation /atmospheric water demand ratio)	The smaller the value, the higher the climate aridity
Root water uptake depth	Depth of root uptake averaged over ten years (2004–2013)	The higher the value, the deeper the depth
Layered SM distribution	Comparing CV of SM between the surface and subsurface	The larger the value, lower variability of deeper soil moisture
Soil water use pattern	Comparing the standardized surface and subsurface dependencies	The higher the value, the higher the surface dependence and vice versa

provides surface soil moisture (0–10 cm) and root zone soil moisture (10–100 cm) products on a $0.25^\circ \times 0.25^\circ$ regular grid with a daily temporal resolution, respectively (Martens et al., 2017). At the same time, we need to acknowledge that this dataset is not perfect and there are also limitations. The surface soil layer of GLEAM is constrained within 10 cm of the surface, and the subsurface layer that we proposed can indicate different depths for different places; however, the limitations of the GLEAM dataset make it impossible for us to accurately quantify the effects of water supply below 1 m. To minimize the impact of this limitation, we chose a quantitative method for determining soil water use patterns by comparing the standardized surface and subsurface dependencies. Because the deeper soil water supply use even below 1 m also occurs in the subsurface layer rather than in the surface layer in the two-layer pattern we proposed in this study, there is a greater subsurface dependence and lower surface dependence. Furthermore, whether satellite measured soil moisture or model estimated soil moisture will inevitably deviate from situ measured soil moisture, which is unavoidable in this study.

2.3. Defining soil water use patterns by the effects of soil moisture across layers on vegetation growth

In this step, we aimed to infer layered soil moisture availability to support vegetation growth by modeling the relationship between

vegetation growth and layered soil moisture use. First, to remove the interference of temperature fluctuation effects on vegetation growth, we fitted a linear regression between temperature and NDVI anomalies in each grid and then used the fitted regression to remove the temperature-driven component (whether positive or negative anomalies) from the NDVI values. Then, we proposed a preset pattern of vegetation growth supported by the water supply from two soil layers (surface and subsurface) and established the relationships between the calibrated NDVI (after removing the temperature impacts) and the layered soil moisture in each grid. As with all analyses of global soil moisture availability for vegetation growth, the covariance of surface soil moisture and subsurface soil moisture is an important consideration, potentially affecting the specific model parameter estimates; thus, these findings should be interpreted with caution. To address this issue, we used a stepwise multiple mixed regression model to select the most important candidate variables (surface or subsurface soil moisture) for explaining the layered soil moisture availability by minimizing the Akaike information criterion (Maire et al., 2015). In this way, the global partitioning of soil moisture-related controls on vegetation growth significantly varied among the three experiments involving the surface layer, subsurface layer or both layers.

We separated the surface and subsurface soil moisture dependence for vegetation growth following equation (1).

$$NDVI(sm) = a \times (w_1 SM_{surface} + w_2 SM_{subsurface}) + b \quad (1)$$

where $NDVI(sm)$ is the temperature-free NDVI at a given surface soil moisture ($SM_{surface}$) and a given subsurface soil moisture ($SM_{subsurface}$), and the optimal values of the corresponding parameters (w_1 and w_2) should be normalized between 0 and 1 for comparability and $w_1 + w_2 = 1$. a is a parameter describing the spread of the response pattern to characterize the water cost for vegetation growth in each grid, and b is the residual value. For the given soil moisture of a grid, the different contributions of surface and subsurface soil moisture to vegetation growth are determined by the differences in the values of parameter w between the surface soil and subsurface soil. The greater the parameter w value of a certain layer is, the greater the vegetation growth dependence on soil moisture in this layer.

The normalized parameters w_1 and w_2 in equation (1) were designed to distinguish surface soil moisture sensitivity and subsurface soil moisture sensitivity and to reveal the different contributions to support vegetation growth between surface and subsurface soil moisture. Based

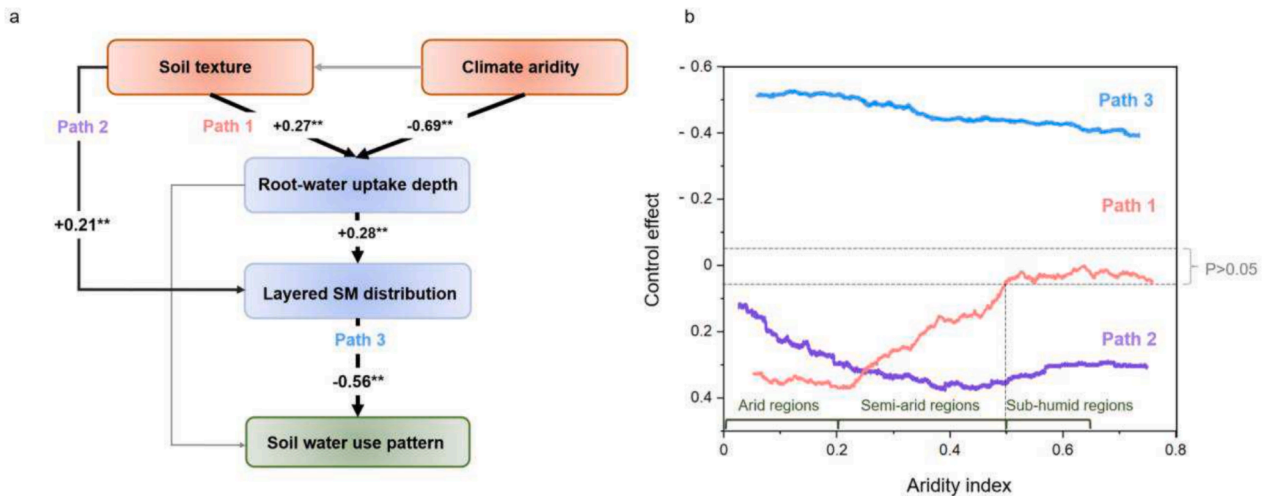


Fig. 5. The mechanisms of climate aridity, soil texture and root depth characteristics determine whether ecological drought is dominated by the surface or subsurface soil layer. (a) A structural equation model (SEM) used to illustrate the plausible interaction pathways of soil texture, root water uptake depth, soil water distribution and soil water use pattern. Solid black arrows represent significant pathways, and gray lines indicate nonsignificant pathways. The numbers near the lines indicate the standard path coefficients, and ** represents P values less than 0.01. (b) Comparison of the three pathways of the controlling effects of soil texture on the soil water use regime for vegetation growth under various climate aridity. Table 1 is a table about the related variables to help understanding Fig. 5a.

on this, we can quantify the layered soil water use pattern by the difference between w_1 and w_2 following equation (2).

$$\text{Layered soil water use pattern} = w_1 - w_2 \quad (2)$$

The closer the difference is to -1 , the greater the dependence of vegetation growth on subsurface soil moisture, and the ecological drought process is dominated by soil moisture fluctuations in the subsurface soil layer. Similarly, the closer the difference is to 1 , the greater the dependence of vegetation growth on surface moisture, and surface moisture variation drives the process of ecological drought.

2.4. Establishment of a potential mechanism that determines soil water use patterns

There is limited opportunity to test soil effects on varied vegetation dependence on surface or subsurface soil moisture across a wider diversity of biomes and climates, especially at spatiotemporal scales of interest to Earth system models. To further explore whether, as we hypothesized, there was a potential relationship between soil layer characteristics and layered soil moisture availability for vegetation growth at a global scale, we used a structural equation model (SEM) to examine the direct and latent relationships between variables, including climate aridity, soil texture, root depth and soil water distribution, that jointly determine layered soil water use patterns for global vegetation. Soil water use patterns revealing water stress from the surface or subsurface soil layer can be designed as a conceptual function involving multiple regulating pathways that occur in soil layers, four direct pathways of climate aridity, soil texture and root-water uptake depth and layered soil water distribution, and many potential indirect pathways, including the indirect effects of soil texture by impacting soil water distribution and root depth, the indirect effects of climate on soil texture and root depth and the effects of root-water uptake depth on soil water distribution.

The aridity index was selected as a climate aridity indicator. For soil texture, we extracted the sand/silt ratio to characterize soil coarsening. ANOVA was also used to determine the differences in soil texture characteristics under the different soil water use patterns. In this study, we also used the soil moisture difference between the surface and subsurface to comprehensively characterize the soil water supply across layers after redistribution (i.e., the layered SM distribution via SEM).

The selected relevant indicators and the quantified soil water use patterns were adopted to construct the structural equation model (SEM). A reasonable SEM should start with a conceptual model with all possible paths and then use a stepwise procedure to remove the paths that did not contribute to the model fit, which can be inferred by the parameters. All plausible interaction pathways derived directly from our theory are shown in the SEMs (Fig. 2). Then, we sequentially eliminated nonsignificant pathways to obtain the parsimonious set of models. Removing those nonsignificant pathways did not improve the model or reduce the goodness of fit. Moreover, the obtained structural equation models based on our hypothesis met the following criteria: the p values of χ^2 and the goodness of fit test $p > 0.05$, the comparative fit index > 0.9 , and the root mean square error of approximation < 0.05 , which all indicated that the model had an excellent fit (Rosseel, 2012). An excellent-fitting SEM helps to further clarify the soil water use regime for vegetation while accounting for the role of climate, soil texture, root depth and soil water distribution across layers.

3. Results

3.1. Global soil water use patterns for vegetation growth

Only 15 % of the grids followed a significant soil moisture use pattern: either surface or subsurface soil moisture controlled vegetation growth, accounting for 7 % and 8 %, respectively, and more than 54 % of global terrestrial ecosystems were supported by both surface and

subsurface soil water supplies. Other areas (approximately 16 %) were farmland areas, urban coverage areas and missing data areas that were excluded from the results to reduce interference from land cover data. There are also some places accounting for approximately 13 % where our method does not work and that do not show clear patterns of vegetation soil water use. They are light yellow, and the value is close to 0 (Fig. 3). These grids are mainly concentrated in tropical and subtropical moist forests, including those in the Amazon region, central Africa and southern China.

The change in climatic aridity did not determine whether vegetation growth was dominated by water from the surface or the subsurface soil layer. Specifically, the subsurface soil water supply capacity dominated vegetation growth in water-limited and humid regions, accounting for 58 % and 60 % of areas experienced subsurface drought stress, respectively. The highest subsoil moisture dependence, indicating a dominant stress from dryness of the subsurface soil layer, limits vegetation growth and is mostly concentrated in southwestern North America, southern Africa and eastern Australia. Moreover, the effects of surface soil moisture on vegetation growth are much smaller in these areas. However, in northern Africa, western Asia and northern China, our result shows that given lower variability of deeper soil moisture in these areas, the survival of vegetation is more reliant on surface soil moisture availability rather than subsurface soil moisture variation.

3.2. Soil texture and water supply under different soil water use patterns

Taking climate aridity as the dividing line, the soil texture characteristics under the different soil water use patterns were calculated separately. The soil texture characteristics were quantified by the ratio of the sand/silt fraction; the higher the score of the ratio fraction was, the coarser the soil texture. There was no significant difference in the sand/silt fraction between water-limited and humid ecosystems. Moreover, the soil texture significantly differed under the different soil water use patterns. In both water-limited regions and humid regions, a coarser soil texture tended to make vegetation growth more dependent on subsurface soil moisture (Fig. 4a).

Based on the plausible pathways proposed by our previous assumption, we further examined whether soil water is separated into different layers with depth to support vegetation growth by establishing the relationships between layered soil water distribution and vegetation soil water use patterns. The difference in the soil water distribution characteristics explained the differences in the vegetation growth dependence on surface and subsurface soil moisture. A greater dependence on subsurface soil moisture for vegetation growth corresponded to higher subsurface soil moisture and a lower coefficient of variation (CV) of soil moisture change relative to the surface soil water supply (Fig. 4b).

3.3. Paths regulating layered soil water use patterns by vegetation

There was a separation of soil water into different layers with depth to support vegetation growth, but the effects of climate aridity, soil texture and root-water uptake depth on vegetation soil water use patterns were achieved via different pathways. There was a direct significant positive effect of an increased sand/silt ratio, resulting in a greater soil moisture supply in the subsurface soil layer than in the surface soil layer, further increasing the dependence of vegetation growth on subsurface soil moisture, which was revealed by the significant correlations in the SEM (Fig. 5a). The relationships between climate aridity and soil water use patterns were mainly connected by affecting the root depth distribution, revealing that increased climate aridity encouraged a deep root depth. However, the root depth was not only regulated by climate but also regulated by soil texture and affected the layered soil water distribution, showing that coarse-textured soil not only enhanced deepwater uptake by roots but also contributed to a higher and more stable subsurface soil water supply relative to the surface soil water supply. This result relied on soil moisture dataset to provide accurate

estimates of surface and subsurface soil moisture. The limitations of the GLEAM (Section 2.2) may impact our results, especially the 10 cm threshold to distinguish surface soil moisture and subsurface soil moisture.

The SEM fitting results clearly indicated that the impacts of soil texture on changing the soil moisture use pattern were exerted by both regulating the layered soil water distribution and root-water uptake depth. The main controlling effect of the soil texture that regulates the soil water distribution represented by Path 2 and Path 3 in the above SEM was always significant ($P < 0.05$), with varied climate aridity gradients represented by the aridity index, revealing that a decrease in the sand/silt ratio was always conducive to increasing the subsoil water supply and thus increasing the dependence of vegetation growth on subsoil moisture across climate zones (Fig. 5b). The process by which the layered soil water distribution regulated by soil texture could determine whether ecological drought was mostly dominated by surface or subsurface soil moisture supply variation was prevalent, and this process peaked in global semiarid regions (Path 2 in Fig. 5b). A coarser soil texture contributed to a higher and more stable subsurface soil water supply relative to the surface soil water supply, regardless of climatic aridity gradients. In contrast to the controlling effect of soil texture on soil water distribution, the effect of the coarse soil texture that enhanced the deep root depth represented by Path 1 in the above SEM peaked in global arid regions but was not always stable with varied climate aridity (Fig. 5a and 5b). This effect did not occur based on the relief of climate aridity. This controlling effect of soil texture was widespread when the aridity index was lower than 0.5, which was approximately the threshold of the global semiarid and humid regions.

4. Discussion and conclusion

In this study, we redefined the soil water use pattern for vegetation growth by quantifying vegetation dependence on layered soil moisture. We distinguished the two major ecosystem-level soil moisture use patterns for vegetation growth from previous studies (Reynolds et al., 2004; Luo et al., 2022; Li et al., 2022b), depending on surface soil moisture or subsurface soil moisture, using full satellite records of long-term vegetation and soil moisture satellite data. In contrast to previous studies in which vegetation growth in water-limited regions was supported by a shallow water pool in the surface soil layer and vegetation growth in humid regions was supported by a more constant deepwater pool in the subsoil layer (Schwinning, 2010; Wang et al., 2021), our results showed that the change in climatic aridity does not determine whether vegetation growth is dominated by water from the surface or subsurface soil layer. Moreover, we also found our method failed to show clear patterns in soil water use pattern for vegetation in tropical or subtropical moist forests, either because of limitations with the GLEAM dataset or because of fundamental difficulties in measuring vegetation dynamics using the NDVI in tropical or subtropical moist forests (Camps-Valls et al., 2021).

Sand, clay and silt fractions were used to examine their effects on soil moisture availability for vegetation growth, but the related findings were highly contradictory. Previous studies have highlighted the local relevance of the silt fraction to ecosystems, as silt adjusts the water holding capacity of soil, and the unit cost of water for vegetation growth decreases as the silt content of soil increases (Paillassa et al., 2020), while other studies stress the importance of the sand fraction to increase the soil moisture availability for vegetation growth by facilitating infiltration and percolation into deeper soil layers and reducing bare soil evaporation (Levine et al., 2016; Zhu et al., 2022). Our results showed that an increased sand/silt ratio contributes to subsurface soil moisture availability for vegetation growth at the global scale, regardless of climate aridity. The effect on soil texture effect caused by impacting soil water distribution is stable under various climate aridity gradients, but the effect of soil texture caused by impacting root distribution becomes weaker or disappears with the release of climate drought stress, which is also supported by observed results from site-scale or regional-scale

studies (Renne et al., 2019; Jiang et al., 2022). Our results revealed critical pathways regulating layered soil water use in soil and provided a missing empirical baseline for divergent ecological drought sensitivities across soil layers.

The main source of uncertainty in this study might come from the satellite-based products used in the analysis regarding their retrieval uncertainties, especially for subsoil moisture, as this layer does not have direct satellite data for assimilation. We reduced the impact of dataset uncertainty by selecting a dataset derived semiempirically and independent of assimilated reanalysis products that applied process-based models. We further note that other water sources, such as bedrock water and groundwater, also impact vegetation productivity, but we did not explore these sources in this study (McCormick et al., 2021; Miguez-Macho and Fan, 2021). Previous studies have shown that vegetation responses to deep soil water sources, including groundwater, can also be largely captured by subsurface moisture dynamics in many regions (Wahr et al., 2004; Li et al., 2022c). Therefore, if vegetation growth is dependent on deeper soil water sources, it is also most likely to be more dependent on the water supply from the subsurface soil layer than on that from the surface layer. Although we focused only on the comparison of surface and subsurface layers when defining soil water use patterns, these omissions did not have a subversive impact on our conclusions.

Despite these uncertainties that cannot be eliminated, our results revealed that the combined impacts of climate, soil texture and root depth lead to divergent soil moisture accessibility and availability for vegetation growth and provided evidence to support the dominant role of soil in determining the soil water use regime. More explicitly, assessing the critical paths that occur in soil layers to determine layered soil water use for vegetation growth at a global scale in the future will benefit a full and detailed understanding of water use and constraints in terrestrial ecosystems.

CRediT authorship contribution statement

Xinrong Zhu: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Lu Wang:** Writing – review & editing. **Liang Shi:** Writing – review & editing. **Lu Wu:** Writing – review & editing. **Boyi Liang:** Writing – review & editing. **Hongyan Liu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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